Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/03/2005
PI Name:	Tittel, Frank K Ph.D.		
Project Title:	Development of Advanced Mid-Infrared Laser Based Gas Sensor Technology		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kosterev, Anatoliy Ph.D. (Rice University) Bakhirkin, Yury Ph.D. (Rice University) Wysocki, Gerard P. Ph.D. (Rice University)		
Grant/Contract No.:	NAG-9-1482		
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Task Description:	This project is a joint Rice University- George Washington University-JPL research and development program of mid-infrared diode laser sources in the 3 to 4.4 micron region and their use in trace gas sensing for NASA applications relevant to spacecraft environmental monitoring and advanced life support. At Rice the primary emphasis will be to demonstrate and emphasize a novel optical detection technique based on an ultra-compact photoacoustic analyzer design that will significantly enhance the performance and robustness of near infrared and mid-infrared gas sensors. The group of Prof. Houston Miller is providing expertise in spectroscopic strategy for detecting Code M molecular target gases as well as developing a cavity ringdown spectroscopy (CRDS) based sensor technology. JPL with semiconductor facilities and the necessary expertise is developing Sb-based diode and type II interband cascade lasers with distributed feedback structures for meeting the spectroscopic application needs. The primary emphasis of this task is the development of		

Task Description:	compact, robust trace gas sensor technology based on quartz enhanced photoacoustic spectroscopy that can be applied to the sensitive, selective and real-time detection of Code M molecular targets (formaldehyde, acetaldehyde, ammonia, carbon monoxide, hydrogen cyanide, carbon dioxide, nitrogene dioxide, HF, acrolein, and water vapor) at specified detection limits for each of the target gases. Manned space missions planned for this century will include longer stays on the International Space Station, expeditions to Mars, as well as return trips to the moon. Hence there is a need to create and maintain an environment suitable for human habitation. This will require the development of instrumentation capable of precise detection, quantifying and controlling a number of molecular trace gas species that are relevant to the air quality of spacecraft habitats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Compact, sensitive and selective gas sensors based on near infrared and mid infrared semiconductor lasers and quartz enhanced photoacoustic spectroscopy. This technology is applicable to atmospheric chemistry, environmental monitoring, chemical analysis and industrial prcess control as well as medical diagnostics based on breath analysis.
Task Progress:	1. Ultrasensitive gas detection by quartz-enhanced photoacoustic spectroscopy in the fundamental molecular absorption bands region. A trace gas sensor based on quartz-enhanced photoacoustic spectroscopy (QEPAS) with a quantum cascade laser operating at 4.55 µm as an excitation source was developed. The sensor performance was evaluated for the detection of N2O and CO. A noise-equivalent (1s) sensitivity of 4 ppbv N2O with 3 s response time to (1-1/e) of the steady-state level was demonstrated [Ref. 1]. The influence of the relevant energy transfer processes on the detection limits was analyzed. This work demonstrated that the QEPAS technique can be successfully used for ultra-sensitive detection of trace gas species excited to v = 1 vibrational levels. The realized sensitivity to absorption D(N2O) = 1.5×10[-8] cm[-1]W/sqrtHz allowed the detection of N2O in air and nitrogen samples with a 4 ppbv noise-equivalent (1s) sensitivity and 3 s response time to (1-1/e) of the steady-state level. However, the results of our experiments indicate thatV-T energy transfer from v = 1 excited level of simple molecules is slower than the relaxation of overtone vibrations [Ref. 2, 3]. In some cases the relaxation rate can be enhanced by adding an appropriate species to the gas sample (e.g., SF6 added to N2 for N2O detection). The vibrational relaxation of CO was found to be too slow to produce a measurable QEPAS signal. Nevertheless, we were able to detect this molecule with 280 ppbv noise-equivalent sensitivity due to the rotational relaxation related signal. Thus, a sensitivity improvement of only a fewtimes (via higher laser power or by the means as discussed in the following paragraph) will make the QEPAS sensor reported in this work useful for atmospheric CO monitoring. SF6 does not significantly promote vibrational deexcitation of CO, although it is efficient in case of N2O. This suggests the involvement of lower vibrational levels of the N2O molecule in V-T energy transfer. We expect that refinement of the microresonator design and fabrication technology will lead to a considerable gain in QEPAS sensitivity. Future designs will minimize the openings at the junctions of the resonator tubes and tuning fork (TF), which will ensure better confinement of the photoacoustic sound waves. Another approach to better QEPAS detection sensitivity is to develop a custom-made TF with lower resonant frequency. The desired frequency shift can be achieved by adding compact weights to the ends of a standard 32.8 kHz TF. 2. Further developments of chemical sensors based on quartz enhanced photoacoustic spectroscopy (QEPAS) Details of the fundamentals of QEPAS, the resonant properties and noise characteristics of quartz tuning forks, a comparison of QEPAS based chemical sensors with conventional PAS sensors are described in Ref. 4. Previously published results for QEPAS based chemical gas sensing are summarized. The to date demonstrated NH3 sensitivity of 5.4 x 10[-9] cm[-1] W/sqrtHz was compared to recent published results of photoacoustic gas sensing by other research groups. An experimental study of the long-term stability of a QEPAS-based ammonia sensor indicated that such a sensor exhibits very low drift, which allows data averaging over 3 hours of continuous concentration measurements. The architecture and implementation of autonomous QEPAS-sensor controller electronics and future developments of QEPAS technique are also outlined in Ref. 4. References: 1. A. A. Kosterev, Y. A. Bakhirkin, R. F. Curl, and F. K. Tittel, “Quartz-Enhanced Photoacoustic Spectroscopy,” Optics Letters 27, 1902-1904 (2002). 2. A. A. Kosterev and F.K. Tittel, “Ammonia detection by use of quartz-enhanced photoacoustic spectroscopy with a near-IR telecommunication diode laser”, Applied Optics, 43, 6213-6217 (2004). 3. D. Weidmann, A.A. Kosterev, F.K. Tittel, N. Ryan, D. McDonald, “Application of widely electrically tunable diode laser to chemical gas sensing with quartz-enhanced photoacoustic spectroscopy”, Optics Letters 29, 1837-1839 (2004). 4. A. A. Kosterev, F.K. Tittel, D. Serebryakov, A. Malinovsky and A. Morozov, “Applications of Quartz Tuning Fork in Spectroscopic Gas Sensing”, Journal of Scientific Instruments Review 76, 043105 (2005)
Bibliography Type:	Description: (Last Updated: 04/30/2007)
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